

Ideation Phase

Define the Problem Statements

Date	30-06-2026
Team ID	23MH1A4257
Project Name	Credit Card Approval Prediction
Maximum Marks	2 Marks

Customer Problem Statement:

In the modern financial sector, credit cards have become an indispensable tool for consumer transactions and short-term financing. For commercial banks and lending institutions, managing the credit application pipeline efficiently is paramount to maintaining market competitiveness and operational agility. Traditional manual underwriting methods, which rely heavily on human loan officers meticulously reviewing each application, are no longer viable. They are inherently slow, prone to subjective bias, and cost-prohibitive when scaling up to handle the massive volume of modern consumer requests, leading to customer friction and lost revenue opportunities.

Furthermore, banks face a critical balancing act between maximizing credit distribution to profitable customers and mitigating the risk of financial default. Approving credit cards for high-risk applicants directly results in substantial financial losses through bad debt and defaults. Conversely, overly conservative approval algorithms or human bottlenecks mistakenly reject creditworthy applicants, which damages customer loyalty and forfeits lifetime customer value. Financial institutions must navigate this tradeoff by evaluating complex, multi-dimensional applicant data, such as income streams, employment stability, debt-to-income ratios, and historical payment behaviors, to predict creditworthiness accurately.

To address these operational and financial challenges, this project aims to develop a robust, data-driven automated credit card approval prediction system utilizing machine learning. By analyzing historical application data and applicant profiles, the predictive model will automatically evaluate incoming applications to classify candidates into "approved" or "rejected" categories. This automated solution will significantly accelerate application processing times, reduce operational overhead, eliminate human bias, and provide highly precise risk assessments, thereby allowing financial institutions to optimize their credit portfolios safely.



Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	I am: A	I'm trying to:	But: I	Because:	Which makes me feel:

	recent college graduate and first-time credit applicant.	Obtain my first credit card to start building a healthy credit history for the future.	received an immediate, automated rejection without any clear explanation.	The bank's risk assessment model flags a lack of historical credit data ("thin file") as a high default risk, failing to account for my stable new income.	Discouraged, confused, and locked out of the financial system.
PS-2	I am: A self-employed graphic designer with a strong, established career but fluctuating monthly income.	I'm trying to: Apply for a premium credit card to separate my business expenses and maximize cash-back rewards.	But: My application is caught in a long manual review loop and ultimately rejected.	Because: The automated underwriting system requires traditional, steady pay stubs and penalizes irregular deposit patterns, incorrectly classifying me as financially unstable.	Which makes me feel: Frustrated, undervalued as a loyal bank customer, and un-supported by modern banking tools.